Global Value Chains, Intellectual Property, and Knowledge Diffusion in Pharmaceuticals

Rethinking the TRIPS Waiver Debate

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A Global Pandemic

- WHO declared COVID-19 a pandemic on **March 11, 2020**.
- A reflexive turn against IP. Early calls to suspend IP **before any vaccines**:
  - medical equipment and ventilators (March 2020)
  - N95 respirators (April 2020)
- Working presumption: IP is a hindrance to supply in an emergency.

- The TRIPS Waiver Proposal
  - **October 2020**: waive core IP for COVID-19 medical products (patents, trade secrets, industrial designs, copyright).
  - **June 2022**: the adopted decision was narrower, covering patents and vaccines only.
  - No member ever invoked it.

Overview & Argument in Brief

Revisit IP waiver debate through the lens of **global value chains (GVCs)**: who makes what inside or outside firm boundaries, and where bottlenecks occurred.

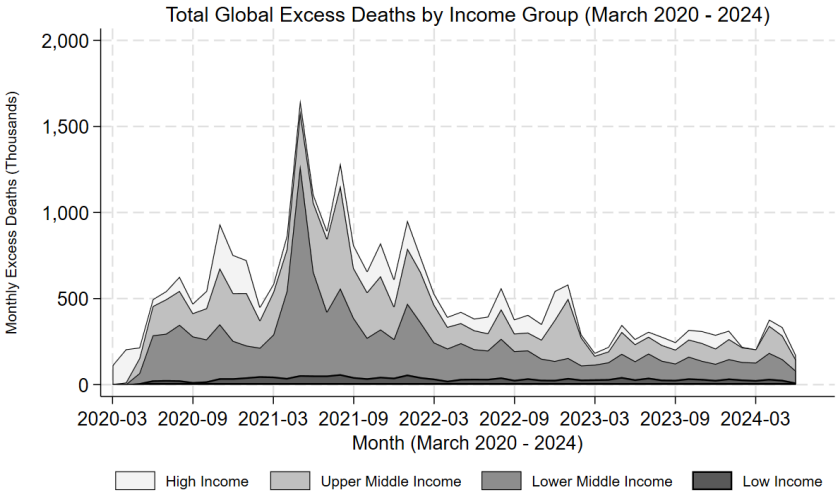
A waiver would not have helped...

- Constraints lay in capacity, inputs, delivery, & coordination
- In the short run, new entrants compete for the same scarce inputs
- Licensees with full IP access still could not scale

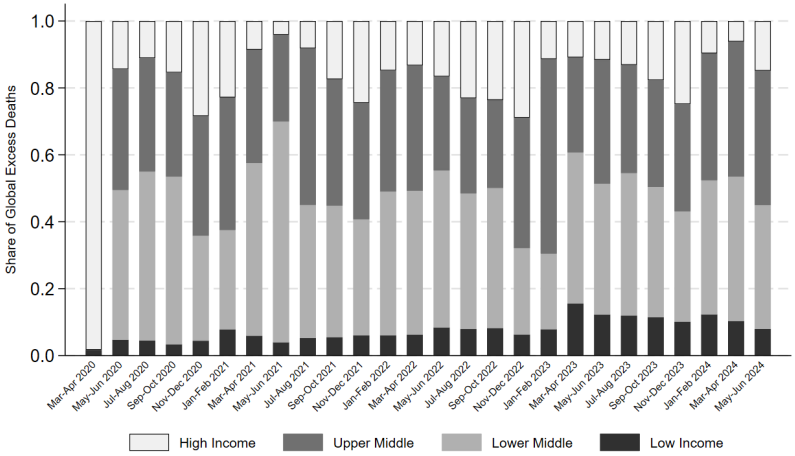
...and better options existed

- The failure was in **ex-ante contract design**
- Access conditions belong in funding agreements up front
- The 2025 Pandemic Agreement is a good start (with provisions on **GVCs**)

Global Excess Deaths, by Income Group



Shares of Global Excess Deaths, by Income Group



Sources: Authors' calculations (OWID)

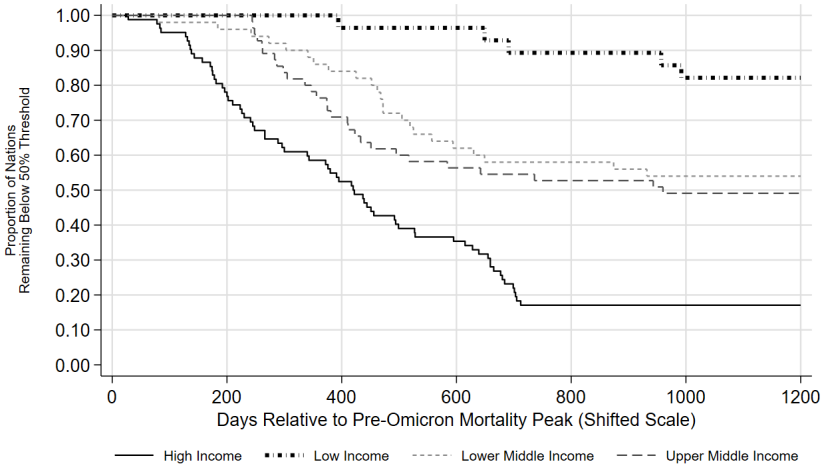
Vaccination Trends

Median Date of Reaching Vaccination Milestones, by Income Group

Group	10%	25%	33%	50%
High Income	May 7, 2021 (73)	Jun 17, 2021 (73)	Jul 1, 2021 (73)	Aug 6, 2021 (69)
Upper Middle	Jul 15, 2021 (53)	Sept 4, 2021 (48)	Oct 1, 2021 (43)	Nov 25, 2021 (28)
Lower Middle	Sept 27, 2021 (47)	Nov 28, 2021 (41)	Jan 15, 2022 (37)	Jan 26, 2022 (23)
Low Income	Mar 6, 2022 (23)	Aug 28, 2022 (15)	Oct 30, 2022 (11)	Sept 4, 2022 (5)

Number of countries in parentheses. Source: Authors' calculations (OWID).

How Long to Reach 50% Full Vaccination?



Source: Authors' calculations, Kaplan-Meier estimates.

Who Ever Reached the Milestones

	Reached 33%	Reached 50%	Countries (pop. share)
High income	98.7%	93.2%	74 (16%)
Upper middle	81.1%	52.8%	53 (37.3%)
Lower middle	75.5%	46.9%	49 (38.6%)
Low income	40.7%	18.5%	27 (8.1%)
Major producers	100%	**93.8%**	16 (53%)
All others	79.1%	58.8%	187 (47%)

- 5.3 billion fully vaccinated, about **two-thirds** of the world, but access tracked income and manufacturing status.

Notes: Full vaccination as of August 12, 2024. Producers per WTO/IMF Covid-19 Vaccine Trade Tracker.

The Case for the Waiver

- **Emergency:** no time to wait for generic or biosimilar entry
- **Monopoly:** patent holders charge more and supply less
- **Flexibilities inadequate:** compulsory licensing too slow and narrow
- **Know-how:** trade secrets are a major barrier, hence a comprehensive waiver
- **Public funding:** R&D and academic work paid by taxpayers
- **Double dipping:** APAs and at-risk funding already bore the risk

Why IP Still Mattered

(a) Innovation was ongoing

- mRNA patent publications: **151** (1971–2019) → **908** (2020–24)
- 1,298 vaccine-related patent filings, Jan. 2020 – Sept. 2022

(b) Funding discovery ≠ funding commercialization

- The **Bayh-Dole** logic: without title, no one might fund trials, scale-up, manufacturing
- Pfizer took no U.S. R&D money; APAs pay only on delivery
- APAs guarantee *buyers*, not *exclusivity*: they are national and confer no control elsewhere

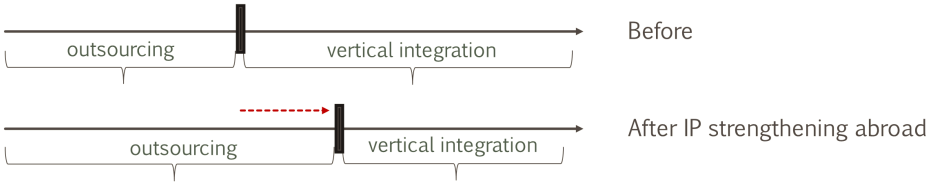
(c) Platform and dynamic incentives

- mRNA was decades of privately funded **platform** IP
- Time-inconsistency: underinvestment in preparedness due to waiver anticipation

(d) IP shapes how value chains are organized

- Not just *how much* innovation, but *who* produces and *where*
- A lead firm decides, task by task, to **integrate** or **outsource**.

GVC Theory and IPR: in Brief



- When stages are **sequential complements**, stronger IP pushes the outsourcing boundary downstream, toward the most knowledge-intensive stages.
- When stages are **sequential substitutes**, IP has a weak influence, as a firm can reduce its dependence on any IP-sensitive input by reallocating tasks across stages.
- Outsourcing and arm's-length licensing matter: they add production capacity that already exists elsewhere and help build absorptive capacity.

Waiver & Know-How

- A waiver removes *legal* barriers; it cannot compel firms to hand over know-how (e.g., cell-culture protocols or process recipes).
- Tacit knowledge transfers through **relationships**: site visits, training, sustained technical support.
- The risk: a trade secret, once disclosed, cannot be recalled. Weaker IP pushes transfers in-house.

Schultz (2024); Ivus, Park & Saggi (2017).

How Long to Form a Chain?

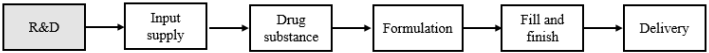
Producer (Day 0)	Drug substance	Input: license	Input: production	Fill and finish
Pfizer/BioNTech (Mar 16, 2020)	+50 (in)	+107 (ext)	+239 (ext)	+50 (in)
Moderna (Jan 23, 2020)	+99 (ext)	–	+126 (ext)	+154 (ext)
AstraZeneca/Oxford (Mar 18, 2020)	+71 (ext)	–	–	+89 (ext)
Johnson & Johnson (Mar 30, 2020)	+2 (in)	–	–	+30 (ext)

- Days from first public milestone; **earliest** documented date, for **experienced firms that already held the rights**.
- An entrant under a waiver starts from scratch: partners, process, its own trials, approval at each site.

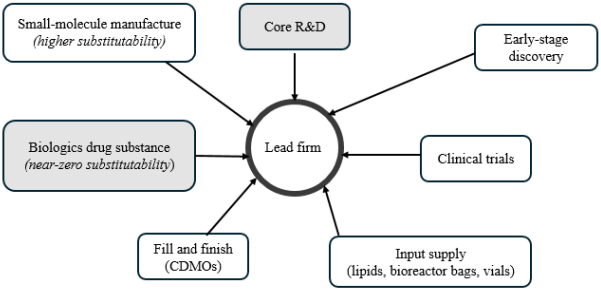
Source: Bown and Bollyky (2022), Appendix. in = in-house; ext = contracted partner.

Snakes, Spiders, and Bottlenecks

(A) Snake: Stages in a single sequence



(B) Spider: Modules feeding assembly in parallel



Source: Authors' illustration. Shaded modules integrated, unshaded outsourced.

Cases

Producer	Arrangement	Constraints
Serum Institute (India)	Voluntary licenses (AstraZeneca, Novavax); CEPI funding	Inputs, engineers, capital; export ban; commitments ahead of capacity
Biological E (India)	Licenses and contracts (J&J)	Capacity and scale-up
Emergent (U.S.)	AstraZeneca and J&J contracts	Contamination
Aspen (South Africa)	J&J fill-finish; license to sell in Africa	No orders: never sold a dose
Afrigen (South Africa)	mRNA from published sources	No know-how; slow scale-up

In these cases, IP wasn't the constraint

What Could Have Been Done Better: Ex-Ante Design

- When governments act as **venture capitalists**, they should write access conditions into funding agreements up front.
- **Contract conditions:** tiered pricing; a non-exclusive license to at least one lower-middle-income manufacturer, *with* know-how.
- **Pre-negotiated pools:** settle licensing terms before a crisis, not during it.
- **A global Operation Warp Speed:** at-risk investment extended internationally.
- **Geographically diversified capacity**, tied to real demand.
- **Upstream input capacity**: lipids, bioreactor bags, filtration units.

The 2025 WHO Pandemic Agreement

Strengths

- Targets the key problems: allocation, capacity, supply-chain resilience
- Reaffirms TRIPS flexibilities; **no waivers**
- Benefit sharing: 20% of real-time production (half donated, half at affordable prices)

Limitations

- Aspirational: core institutions do not yet exist
- Supply-chain network deferred to a future conference; financing unspecified
- Benefit-sharing annex still under negotiation
- "As mutually agreed": largely voluntary

Lessons

- Vaccines met **speed and scale**, but not access.
- The binding constraints were capacity, inputs, demand, and coordination, not IP.
- IP shapes how chains are organized; secure rights support the **outsourcing and licensing** through which know-how reaches the developing world.
- Spider-like chains: coordination, not legal access, was the problem.
- Access is best settled **before** a pandemic, through funding and procurement design.